

Reproduction and Trace-Prefix Validation of SIEVE Cache Eviction

FIFO, LRU, CLOCK, and SIEVE under Controlled Warmup

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Independent academic reproduction

Compute platform: Universitas Brawijaya

Technical Report

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Abstract

This report reproduces the bundled example and validates the SIEVE cache eviction policy using the official NSDI 2024 artifact. FIFO, LRU, CLOCK, and SIEVE are evaluated on three consecutive one-million-request windows from each of Wikimedia CDN, Twitter KV, and Meta CDN trace prefixes. All policies and warmup conditions share an 800,000-request evaluation suffix per window. Object and byte capacity modes, eight capacity fractions, and cold, 10%, and 20% warmup produce 1,728 configurations. At 10% capacity and 20% warmup, SIEVE reduces Wikimedia object and byte miss ratios relative to LRU by 6.43% and 8.99%, respectively. Twitter results are nearly tied, while Meta rankings depend on configuration and metric. Warmup sensitivity remains measurable. These observations support workload-specific benefits, not universal superiority or complete reproduction of the original paper. The archive includes actual measurements, workflow logs, pinned runners, analysis, hashes, and reporting code.

Contents

1. Objective and Experimental Method	3
2. Results	4
3. Warmup Sensitivity and Comparative Curves	5
4. Discussion, Limitations, and Reproduction	6
References and Evidence Provenance	7
Report Documentation Page	8

Abbreviations and Definitions

Term	Meaning
OMR	Object miss ratio: missed requests / evaluated requests.
BMR	Byte miss ratio: bytes requested on misses / total requested bytes.
Working set	Distinct IDs, or sum of maximum observed sizes per ID, in the full window.
Percentage point	Absolute difference between two ratios, multiplied by 100.
Relative reduction	$(\text{LRU ratio} - \text{SIEVE ratio}) / \text{LRU ratio}$, multiplied by 100.

Evidence status: all 1,728 actual result rows passed independent archive checks. The server workflow reports successful completion.

1. Objective and Experimental Method

The objective is to verify that the official SIEVE artifact can run on the available Linux platform and to test whether the example-trace advantage persists on real workload prefixes. The study compares eviction policies; it does not measure serving-system latency or GPU performance. The algorithm and artifact are credited to the original authors [1, 2].

Item	Configuration
Artifact	cacheMon/NSDI24-SIEVE, commit 0861f8261c37b8286a70449a1defb5a4749c72d2.
Workloads	Wikimedia wiki_2019t; Twitter cluster10; Meta meta_reag; oracleGeneral records from CMU [3].
Windows	Three consecutive windows of 1,000,000 nonzero-size requests per trace.
Evaluation	Same last 800,000 requests in each window for every condition.
Warmup	Cold: 0; warm10: preceding 100,000; warm20: preceding 200,000 requests.
Capacities	0.1%, 0.5%, 1%, 2%, 5%, 10%, 20%, and 40% of each window working set.
Policies / modes	FIFO, LRU, CLOCK, SIEVE; object and byte capacity modes.
Execution	Four simulator threads; 54 sequential invocations; 1,728 result rows.

The artifact interface uses time-based warmup. Each prepared scenario therefore uses ordinal request timestamps, making the multi-cache branch consume the exact intended number of warmup requests. IDs, sizes, and request order are preserved. TTL and metadata overhead are disabled; future-access fields are not used by these four policies. Warmup requests affect cache state but are excluded from evaluation denominators.

A private simulator source copy changes reporting only: capacities are printed exactly and ratios use ten decimal places. Eviction code is unchanged. The original source and patch are retained separately. The host exposed 256 CPUs, but its CPU bandwidth quota was equivalent to 16 cores. Approximately 2 TiB of host RAM was reported as shared capacity; no exclusive allocation is claimed. The visible A100 GPU was unused.

2. Results

Table 1 reports the arithmetic mean of three consecutive windows at 10% working-set capacity and 20% warmup. Object mode uses OMR; byte mode uses BMR. Lower is better. Capacity budgets differ between modes and absolute capacities differ between windows.

Trace / mode	FIFO (%)	LRU (%)	CLOCK (%)	SIEVE (%)	Rel. vs LRU (%)
wiki-CDN / objects	75.0703	72.3767	71.3520	67.7217	+6.4316
wiki-CDN / bytes	69.7164	66.2399	64.9621	60.2860	+8.9884
twitter-KV / objects	50.0011	50.0011	50.0011	50.0027	-0.0032
twitter-KV / bytes	50.0011	50.0011	50.0011	50.0024	-0.0026
meta-CDN / objects	43.7267	42.8628	42.7352	42.7063	+0.3652
meta-CDN / bytes	23.4489	23.2817	23.3881	23.4272	-0.6248

Table 1. Miss ratios and SIEVE relative reduction at the selected capacity and warmup. Positive relative reduction favors SIEVE. Relative reduction is computed from the mean ratios, not averaged individual percentages.

SIEVE has the lowest primary miss ratio among all four policies in all 72 Wikimedia configurations per mode. Its Wikimedia advantage at the selected condition is 4.6550 percentage points in object mode, equivalent to 6.4316% relative reduction. Byte mode improves by 5.9539 percentage points, equivalent to 8.9884% relative reduction.

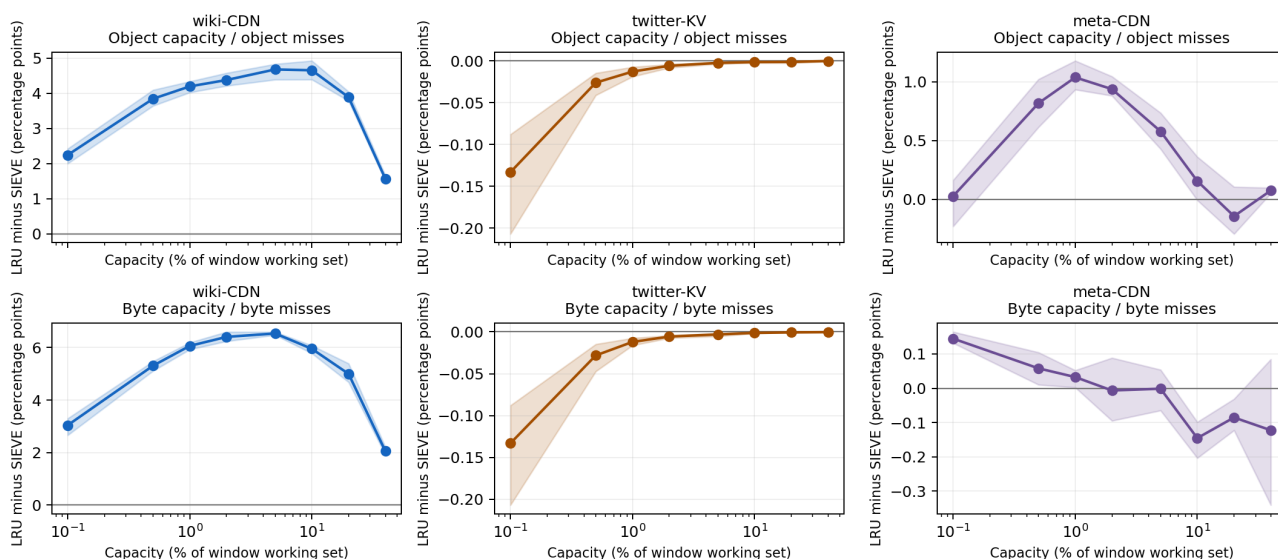
Trace / mode	Wins	Ties	Losses	Lowest of four (incl. ties)
wiki-CDN / objects	72	0	0	72/72
wiki-CDN / bytes	72	0	0	72/72
twitter-KV / objects	0	1	71	1/72
twitter-KV / bytes	0	2	70	2/72
meta-CDN / objects	64	0	8	59/72
meta-CDN / bytes	38	0	34	33/72

Table 2. SIEVE versus LRU over three windows, three warmup conditions, and eight capacities. Counts describe configurations, not independent statistical samples. Twitter differences are small despite the many losses. Meta contains both wins and losses.

3. Warmup Sensitivity and Comparative Curves

For Wikimedia SIEVE, moving from cold cache to 20% warmup lowers the mean primary miss ratio across windows and capacities by 1.6237 percentage points in object mode and 1.9038 in byte mode. Increasing warmup from 10% to 20% still lowers it by 0.5059 and 0.5890 points, respectively. These changes show warmup sensitivity; they do not establish steady state.

SIEVE versus LRU on real trace prefixes — 20% warmup



Positive: SIEVE has fewer misses. Line: mean of 3 consecutive windows. Band: observed min-max, not a confidence interval.

Figure 1. LRU minus SIEVE miss ratio in percentage points under 20% warmup. Positive values favor SIEVE. Lines show means of three consecutive windows; shading shows observed minimum and maximum, not a confidence interval. Panel vertical scales differ.

For Meta, cold-to-20% mean improvements are only 0.2493 points in object mode and 0.0495 in byte mode. Twitter changes are near zero. More warmup can occasionally worsen a particular result because the initial cache state changes; monotonic improvement is not assumed.

4. Discussion, Limitations, and Reproduction

Workload dependence. SIEVE performs consistently well on the tested Wikimedia prefix. Twitter is nearly tied at larger capacities, and SIEVE is slightly worse in most tested configurations. On Meta, SIEVE sometimes wins and sometimes loses. For Meta window 3, byte capacity 40%, warmup 20%, LRU BMR is 19.9362% and SIEVE BMR is 20.2767%, a 0.3405-point disadvantage for SIEVE.

Size and metric interpretation. Comparing object-mode OMR with byte-mode BMR changes both budget and objective. Ranking changes cannot be attributed solely to object-size heterogeneity. An isolating follow-up should compare both metrics at the same byte capacity and replay the same IDs with normalized sizes.

Scope. This study uses only three million nonzero-size requests from each trace prefix. Adjacent windows do not constitute independent random samples. No confidence intervals or significance tests are asserted. No throughput, latency, concurrency-scalability, or GPU claim follows from miss-ratio simulation. This is not a complete reproduction of the paper.

Evidence. The archive contains actual CSV measurements and aggregate server logs. Per-invocation simulator logs, source environment lock files, server manifests, and prepared binary traces remain on the server and were not supplied for this archive. Successful completion indicates the runner checked measured and warmup request counts; independent archive checks verify CSV completeness, metadata, paired capacities, and evidence hashes.

Reproduce on Linux. Install Git, GCC/G++, Make, and Conda. Clone the private repository using an authenticated GitHub session, then execute:

```
bash scripts/bootstrap_artifact.sh
bash scripts/run_zipf.sh
bash scripts/run_realworld.sh
python3 scripts/analyze_results.py
```

The bootstrap pins the artifact commit. Each experiment uses a fresh timestamped result directory. A WINDOW_REQUESTS=10000 pilot validates the pipeline but is not equivalent to the archived full run. Runners require outbound HTTPS and do not need sudo or Docker.

Conclusion. The artifact builds and runs on the available CPU platform. Real-prefix measurements support workload-specific SIEVE benefits, with persistent warmup sensitivity and counterexamples to universal superiority. Longer time coverage, additional clusters, and warmup-convergence checks are appropriate next experiments.

References and Evidence Provenance

[1] Zhang, Y.; Yang, J.; Yue, Y.; Vigfusson, Y.; and Rashmi, K. V. SIEVE is Simpler than LRU: an Efficient Turn-Key Eviction Algorithm for Web Caches. USENIX NSDI, 2024. <https://www.usenix.org/conference/nsdi24/presentation/zhang-yazhuo>

[2] cacheMon. NSDI24-SIEVE: official artifact. <https://github.com/cacheMon/NSDI24-SIEVE>. Commit 0861f8261c37b8286a70449a1defb5a4749c72d2. Apache-2.0.

[3] Carnegie Mellon University, Parallel Data Laboratory. Cache datasets linked by the SIEVE artifact. <https://ftp.pdl.cmu.edu/pub/datasets/twemcacheWorkload/cacheDatasets/>

[4] NASA. NPR 2200.2E, Requirements for Documentation, Approval and Dissemination of Scientific and Technical Information, revalidated with Change 2. <https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=2200&s=2E>. Accessed 9 October 2026.

[5] U.S. General Services Administration. Standard Form 298, Report Documentation Page, revision May 2020. <https://www.gsa.gov/reference/forms/report-documentation-page>. Accessed 9 October 2026.

References [4, 5] guide report organization and metadata. Agency-specific approval workflows, logos, sponsorship claims, official report numbers, and government distribution codes are not adopted. This independent report makes no claim of NASA certification or full agency publication compliance.

Archive Provenance

Real-world run: realworld-20261008T233726Z-589911

Zipf run: 20261008T210208Z-567188

The original bundled Zipf run completed four policies across eight capacities on one million requests. At capacity 100 objects, SIEVE OMR was 32.88% and LRU OMR was 42.47%; these example results motivated real-trace validation.

results/realworld/miss_ratios.csv
6c586195a31da3b0d5bdf4b27338089f
de1c26c0cddcc41e70eacff866df4e35

results/realworld/workflow.log
1ab640bb5a7cbf83ac55df9b23863411
6f9428083d7f9e6d0df3bf4996a5ff11

results/zipf/workflow.log
44868901d43da97b743580a3a222d447
aa10002a99bf1cfe2c7aeb5ce04cb97d

Report Documentation Page

Adapted from SF 298 metadata categories [5] for this independent academic report. This page is not a completed government submission form.

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6. Author	Zakaria Rafi
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8. Report number	ZR-SIEVE-TR-2026-001 (author-assigned)
9-11. Sponsor / monitor / numbers	Not applicable; no NASA or government sponsorship asserted.
12. Distribution / availability	Private GitHub repository; access controlled by the author. No government distribution code assigned.
13. Supplementary notes	NASA STI-structured academic report; official SIEVE artifact pinned; source CSV/logs included.
14. Abstract	This report reproduces the bundled example and validates the SIEVE cache eviction policy using the official NSDI 2024 artifact. FIFO, LRU, CLOCK, and SIEVE are evaluated on three consecutive one-million-request windows from each of Wikimedia CDN, Twitter KV, and Meta CDN trace prefixes. All policies and warmup conditions share an 800,000-request evaluation suffix per window. Object and byte capacity modes, eight capacity fractions, and cold, 10%, and 20% warmup produce 1,728 configurations. At 10% capacity and 20% warmup, SIEVE reduces Wikimedia object and byte miss ratios relative to LRU by 6.43% and 8.99%, respectively. Twitter results are nearly tied, while Meta rankings depend on configuration and metric. Warmup sensitivity remains measurable. These observations support workload-specific benefits, not universal superiority or complete reproduction of the original paper. The archive includes actual measurements, workflow logs, pinned runners, analysis, hashes, and reporting code.
15. Subject terms	Cache eviction; SIEVE; LRU; CLOCK; FIFO; trace replay; warmup; reproducibility
16. Security classification	No government security classification asserted. Repository privacy is an access setting.
17. Limitation of abstract	Author-controlled repository access.
18. Number of pages / 19. Contact	8 / GitHub account zakariarafi